



10.3.2 Givens' rotations ¶

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fit width

We now introduce another important class of orthogonal matrices known as Givens' rotations. Actually, we have seen these before, in [Subsubsection 2.2.5.1](#), where we simply called them rotations. It is how they are used that makes them Givens' rotations.

Given a vector $x = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} \in \mathbb{R}^2$, there exists an orthogonal matrix G such that $G^T x = \begin{pmatrix} \pm \|x\|_2 \\ 0 \end{pmatrix}$. The Householder transformation is one example of such a matrix G . An alternative is the Givens' rotation: $G = \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix}$ where $\gamma^2 + \sigma^2 = 1$. (Notice that γ and σ can be thought of as the cosine and sine of an angle.) Then

$$\begin{aligned} G^T G &= \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix}^T \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix} = \begin{pmatrix} \gamma & \sigma \\ -\sigma & \gamma \end{pmatrix} \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix} \\ &= \begin{pmatrix} \gamma^2 + \sigma^2 & -\gamma\sigma + \gamma\sigma \\ \gamma\sigma - \gamma\sigma & \gamma^2 + \sigma^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

which means that a Givens' rotation is an orthogonal matrix.

Homework 10.3.2.1. Propose formulas for γ and σ such that

$$\begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix}^T \underbrace{\begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix}}_x = \begin{pmatrix} \|x\|_2 \\ 0 \end{pmatrix},$$

where $\gamma^2 + \sigma^2 = 1$.

▼ [Solution](#)

Take $\gamma = \chi_1 / \|x\|_2$ and $\sigma = \chi_2 / \|x\|_2$, then $\gamma^2 + \sigma^2 = (\chi_1^2 + \chi_2^2) / \|x\|_2^2 = 1$ and

$$\begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix}^T \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} \gamma & \sigma \\ -\sigma & \gamma \end{pmatrix} \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} (\chi_1^2 + \chi_2^2) / \|x\|_2 \\ (\chi_1 \chi_2 - \chi_1 \chi_2) / \|x\|_2 \end{pmatrix} = \begin{pmatrix} \|x\|_2 \\ 0 \end{pmatrix}.$$

Remark 10.3.2.1. We only discuss real-valued Givens' rotations and how they transform real-valued vectors, since the output of our reduction to tridiagonal form, after postprocessing, yields a real-valued tridiagonal symmetric matrix.

Ponder This 10.3.2.2. One could use 2×2 Householder transformations (reflectors) instead of Givens' rotations. Why is it better to use Givens' rotations in this situation.